

Internship Report – Day 5

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 5 Overview:

We set up an ANPR system in Python using VS Code that detects vehicle number plates from a CCTV video feed, extracts the text using OCR (EasyOCR), and verifies the number against an authorized list to decide whether to open a barricade. The system uses YOLOv8 for real-time plate detection and will be trained on a license plate dataset (from Roboflow or Kaggle) in YOLOv8 format. Once the dataset is downloaded and organized, we'll train a custom model, integrate it into the detection script, and complete the smart parking entry system.

Tools Used Today:

VS CODER, GEMINI, SONNET.

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

Approach to the Task:

I wrote a Python script using OpenCV that opens the webcam, detects the number plate in the video stream, and captures that specific region as a photo. Once the number plate is identified based on its shape and size, the script crops that area from the frame and saves it automatically to a folder. This helps in directly storing the number plate image without saving the entire frame.

Attachments:-

```
simple_anpr.py > ...
1  import cv2
2  import easyocr
3  import numpy as np
4  import os
5  from datetime import datetime
6
7  # Initialize the OCR reader
8  reader = easyocr.Reader(['en'])
9
10 # Open the default camera
11 cap = cv2.VideoCapture(0)
12
13 output_dir = 'anpr_output'
14 os.makedirs(output_dir, exist_ok=True)
15
16 print("Press Ctrl+C in the terminal to stop.")
17
18 frame_count = 0
19 try:
20     while True:
21         ret, frame = cap.read()
22         if not ret:
23             break
24
25         # Optional: Resize for faster processing
26         img = cv2.resize(frame, (640, 480))
27
28         # Run OCR on the frame
29         results = reader.readtext(img)
30         detected = False
31         for (bbox, text, prob) in results:
32             # Draw rectangle around detected text
33             (top_left, top_right, bottom_right, bottom_left) = bbox
34             top_left = tuple(map(int, top_left))
35             bottom_right = tuple(map(int, bottom_right))
36             cv2.rectangle(img, top_left, bottom_right, (0, 255, 0), 2)
37             # Show detected text
38             cv2.putText(img, text, (top_left[0], top_left[1] - 10), cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0), 2)
39             print(f"Detected: {text}")
40             detected = True
41         if detected:
42             timestamp = datetime.now().strftime('%Y%m%d_%H%M%S%f')
43             filename = f"frame_{timestamp}.jpg"
44             filepath = os.path.join(output_dir, filename)
45             cv2.imwrite(filepath, img)
46             print(f"Saved frame with detection to: {filepath}")
47             frame_count += 1
48     except KeyboardInterrupt:
49         print("\nStopped by user.")
50
51 cap.release()
```





